Solid State Optics

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Versuchsdurchführung: ??. ?.????
Protokollabgabe: ?? ??

1 Introduction

Fourier Transform Infrared (FTIR) spectroscopy is a powerful method for determining the electronic properties of metals and semiconductors. The FTIR technique is based on the interaction of infrared light with the material. Reflection and transmission are used to determine important electronic properties. Especially for semiconductors, the FTIR method is particularly useful because it helps to adjust parameters such as the doping concentration or the bandgap. For building applications like LEDs, transistors, or solar cells, precise knowledge of the bandgap and doping concentrations is required. In contrast to other common methods, such as spectroscopy with a monochromator, the FTIR method is much faster and more precise. Instead of varying the wavelength through multiple measurements, the FTIR method can measure the entire spectrum at once due to the use of a Michelson interferometer, which enables the interference of light. After Fourier transformation, the complete spectrum is obtained in a single step.
Firstly, the theoretical basics of light-matter interaction and some important formulas regarding reflection and transmission will be discussed. Following this theoretical introduction, the experimental setup will be described. The final part of this paper will focus on data evaluation, starting with the gas absorption of air. The second part of the data evaluation will involve determining the signal-to-noise ratio. The last part will address the determination of several previously mentioned electronic and optical properties, such as the refractive index, extinction coefficient, absorption coefficient, bandgap, and pulse matrix element for all given semiconductor samples.

2 Theory

2.1 Wave equations

The propagation of electromagnetic waves in a medium are described with the wave equations:

$$\nabla^2 \vec{E} = \frac{\mu\varepsilon}{c^2}\frac{\partial^2 \vec{E}}{\partial t^2} + \sigma\mu_0\mu\frac{\partial \vec{E}}{\partial t} \tag{1}$$

$$\nabla^2 \vec{H} = \frac{\mu\varepsilon}{c^2}\frac{\partial^2 \vec{H}}{\partial t^2} + \sigma\mu_0\mu\frac{\partial \vec{H}}{\partial t} \tag{2}$$

with $\vec{E}$ as the electric field vector, $\vec{H}$ as the magnetic field vector, μ as the permeability, ε as the permittivity, σ as the conductivity, and c as the speed of light in vacuum.
Under the assumption of plane waves with the components $\vec{E} = E_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)}$ and $\vec{H} = H_0 e^{i(\vec{k}\cdot\vec{r}-\omega t)}$ can be used as an ansatz to solve the wave equations 1 and 2. The dispersion relation of the plane waves is then described with:

$$k^2 = \omega^2 \cdot (\epsilon_0 + i\frac{\sigma}{\omega\epsilon_0}) \tag{3}$$

with k as the wave vector, ω as the angular frequency, and ϵ_0 as the permittivity of vacuum.

2.2 Plane parallel layer

The investigated semiconductor samples can in first approximation be described as a plane parallel layer 1 with a thickness d and refractive index N_2 which acts

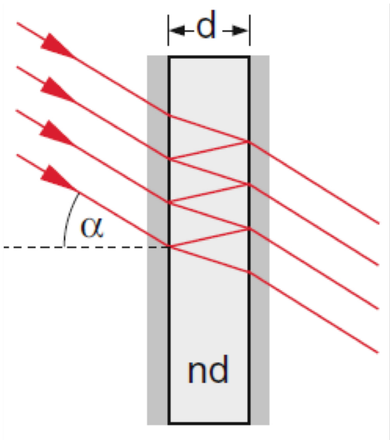

Abbildung 1: Simplified illustration of the measured samples as a plane parallel plate with thickness d and refractive index N_2. The surrounding air is described with a refractive indes $N_1 \approx 1$ [?]

like a fabry-perot interferometer. Light is simplified as beams. Every incoming beam is partially reflected and partially transmittet at the first interface. The total power reflectivity R_1 is described with:

$$R_1 = \left| \frac{1 - N_2}{1 + N_2} \right|^2 \tag{4}$$

without considering the phase shift of the reflected beam. Like shown in fig. 1, the total power reflectivity of R_n then consists of a reflection of the incoming beam at the interfaces of the material. Additionally, the absorption of the material is considered. Therefore, R_n is described with:

$$R_{\mathrm{n}+1} = R_\mathrm{n}(1 - R_1)^2 \cdot e^{-2\beta d} \tag{5}$$

with the absorption coefficient β and the total power reflectivity R_n of the n^{th} interface. The assumption that the reflection of the first interface is the same as the reflection of the second interface is in reality not correct. Disturbances like roughness of the interfaces or the absorption of the material lead to a different reflection coeffitients of the two interfaces.
Considering this and the phase of the light, the total power reflectivity R is described with:

$$R = \left| \frac{(e^{i\frac{2\omega}{c}N_2 d} - 1)(1 - N_2)}{e^{i\frac{2\omega}{c}N_2 d}(1 - N_2) - (1 + N_2)} \right|^2 . \tag{6}$$

ω is the angular frequency of the light and c the speed of light. The phase shift of the light is described with $e^{i\frac{2\omega}{c}N_2 d}$. In contrast to 5, equation 6 shows a periodic modulation of the reflectivity because of interference. The difference between the maxima of the modulation can be calculated with:

$$\Delta\nu = \nu_{\mathrm{m}+1} - \nu_\mathrm{m} = \frac{1}{2dN_2} \tag{7}$$

2.3 Optical properties

The refractive index N:

$$N(\omega) = n(\omega) + i\kappa(\omega) \tag{8}$$

consists of a real part n and an imaginary part κ and is related to the dielectric function ϵ:

$$\epsilon = N^2 = n^2 - \kappa^2 + 2in\kappa \tag{9}$$

2.4 Drude model

The drude model is a classical model in which the electrons in a solid are treated as a gas of free particles. In this model the dielectric function ϵ can be described as:

$$\epsilon = 1 - \frac{\omega_\mathrm{p}^2\tau^2}{1 + \omega^2\tau^2} + i\frac{\omega_\mathrm{p}^2\tau}{\omega(1 + \omega^2\tau^2)} \tag{10}$$

with ω_p as the plasma frequency:

$$\omega_\mathrm{p} = \sqrt{\frac{N_\mathrm{e}e^2}{\epsilon_0 m^*}} \tag{11}$$

and τ as the relaxation time of the electrons. m^* is the effective mass of the electrons and N_e the charge carrier density. At the plasma frequency ω_p the real part of the refractive index becomes nearly zero $n(\omega_p) \approx 0$ and therefore the phase velocity and wavelength diverge. This means that all electrons are oscillating in phase. Absorption of light is not possible for frequencies below the plasma frequency.
Eq. 10 can be simplified through the introduction of the high frequency conductivity σ

$$\sigma = \frac{N_\mathrm{e}e^2\tau}{m^*}\frac{1}{1 - i\omega\tau} \tag{12}$$

to a expression including the lattice contribution of the material ϵ_LC:

$$\epsilon = \epsilon_\mathrm{LC} + i\frac{\sigma}{\omega\epsilon_0}. \tag{13}$$

For metals the lattice contribution is set to one $\epsilon_\mathrm{LC} = 1$ as the electrons are considered to be a free electron gas. Using equations 9,10 and 11 with the effective mass $m^* = m_\mathrm{e}$ equal to the electron mass, the theoretical reflectivity of a metal can be calculated. The result is shown in fig 2. The used constants for the theoretical reflection of a metal are listed in table 2

For frequencies below the plasma frequency the reflectivity is nearly one. However, if the frequency of the light is larger than the plasma frequency, the reflectivity is significantly reduced.

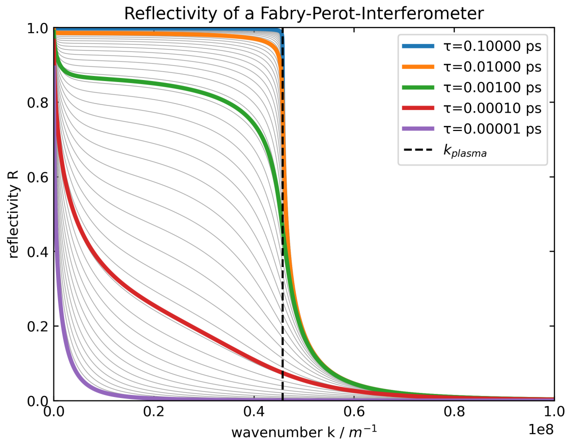

Abbildung 2: Theoretical reflectivity of a metal according to the drude model. The reflectivity is reduced significantly for wavenumbers above k_{plasma}

charge carrier density N_{e}	$5.9 \cdot 10^{28}\,\mathrm{m}^{-3}$
electron mass m_{e}	$9.11 \cdot 10^{-31}\,\mathrm{kg}$
layer thickness d	$500\,\mu\mathrm{m}$

Tabelle 1: Constants used for the calculation of the theoretical reflectivity of a metal, shown in fig 2. The charge carrier Density is taken from [4]

2.5 Semiconductors

While metals are described properly with the drude model, for semiconductors the bandgap has to be considered additionally. Like shown in 3 the bandgap of a semiconductor can be direct or indirect. To ensure energy and momentum conservation, the excitation of a valence electron to the conduction band for a indirect bandgap is only possible with an additional phonon absorption or emission process. Electro magnetic waves are able to polarize the lattice of the semiconductor.¿nahe am AP? This leads to a change in the lattice contribution ϵ_{LC} of the dielectric function:

$$\epsilon_{\mathrm{LC}} = \epsilon_\infty \left(1 + \frac{\omega_{\mathrm{LO}}^2 - \omega_{\mathrm{TO}}^2}{\omega_{\mathrm{TO}}^2 - \omega^2 - i\Gamma\omega}\right) \tag{14}$$

with ϵ_∞ as the high frequency dielectric constant, ω_{LO} as the longitudinal optical phonon frequency, ω_{TO} as the transversal optical phonon frequency and Γ as the damping constant. Together with eq. 13 the theoretical reflectivity of a semiconductor can be calculated as shown in fig. 4.

For high frequencies the characteristic of the band gap needs to be considered, as inter band excitation processes are possible. For a direct bandgap the imaginary part of the dielectric function is described

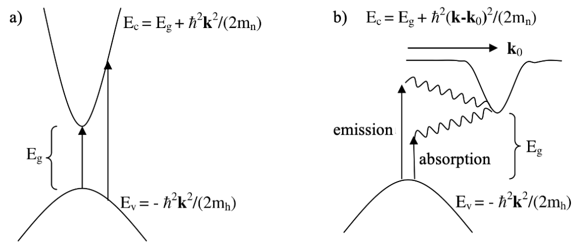

Abbildung 3: a) direct and b) indirect semiconductor bandgap. If a photon is absorbed by a valence electron the excitation to the conduction band is for the indirect bandgap only possible with an additional phonon absorption or emission process.[2]

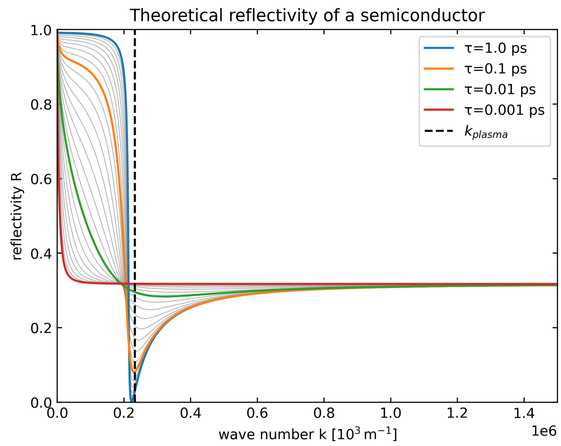

Abbildung 4: Theoretical reflectivity considering the drude model and polarization effects.

with:

$$\epsilon_d'' = \frac{e^2 (2\mu)^{\frac{3}{2}} |p_{\mathrm{cv}}|^2}{h \epsilon_0 m_{\mathrm{e}}^2 (\hbar\omega)^2} \sqrt{\hbar\omega - E_{\mathrm{g}}} \tag{15}$$

and for an indirect bandgap with:

$$\epsilon_{\mathrm{id}}'' \propto (\hbar\omega \pm \hbar\Omega_{\mathrm{ph}} - E_{\mathrm{g}}) \tag{16}$$

for $\hbar\omega \geq E_{\mathrm{g}} \pm \hbar\Omega$. $|p_{\mathrm{cv}}|$ is the pulse matrix element of the interband transition, μ is the reduced mass of the electron-hole pair, E_{g} is the bandgap energy, Ω_{ph} is the phonon resonance frequency and m_{e} is the effective mass of the electron.

3 Set-Up

Fig. 5 shows a simplified schematic version of the *Bruker IFS 66 v/s* fourier spectrometer. The source (1) emmits light in the near- infrared range ($1500\,\mathrm{cm}^{-1}$ – $15000\,\mathrm{cm}^{-1}$) which is then reflected on the beam splitter (3). The Beamsplitter, static mirror (4) and mo-

charge carrier density N_{sc}	$1.05 \cdot 10^{24}\,\mathrm{m}^{-3}$
effective mass m_{eff}	$0.063 \cdot m_e$
layer thickness d	$500\,\mu\mathrm{m}$
damping constant Γ	$250\,\mathrm{m}^{-1}$
l.o. phonon frequency ω_{lo}	$\hbar^{-1}10^{-3} \cdot 36.1\,\mathrm{meV}$
t.o. phonon frequency ω_{to}	$\hbar^{-1}10^{-3} \cdot 33.2\,\mathrm{meV}$
relative dielectric constant ϵ_{inf}	10.89

Tabelle 2: Constants used for the calculation of the theoretical reflectivity of intrinsic GaAs, shown in fig 2. Constants are taken from [5]

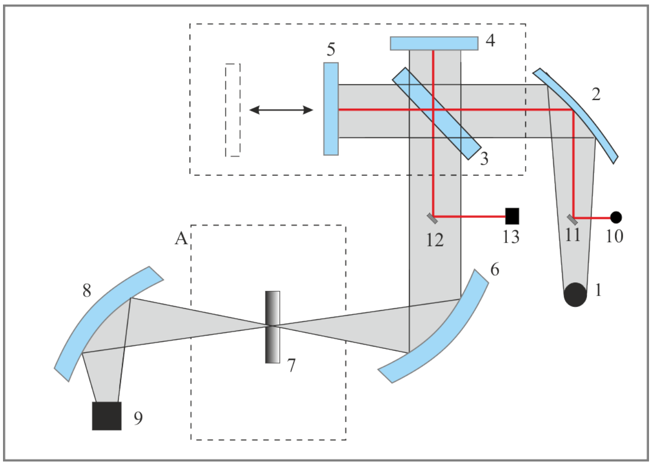

Abbildung 5: Simplified schematic illustration of the set-up used for all experiments. A short overview of the working principle is given in the text.[3]

vable mirror (5) can be seen as a michelson interferometer which is used to create a interference pattern. During the measurement, the movable mirror is moved with constant speed. The light is then focused via another mirror (6) on the sample (7). In this case, transmission is shown. For reflection measurements an additional module is used which is not shown in the figure. Mirror (8) then focuses the light onto the detector (9). The signal is then digitalized. Numbers (10) to (13) show the beampath of a laser which is used to determine the position of the movable mirror.

4 Experiment

All following transmission and reflection experiments are done at ambient air in the chamber. The evaluation of the signal with a 'Mertz' phase correction is done automatically by the system. Furthermore, the spectal range for all measurements is set to: $(1500 - 15000)\,\mathrm{cm}^{-1}$.

4.1 Gas absorption

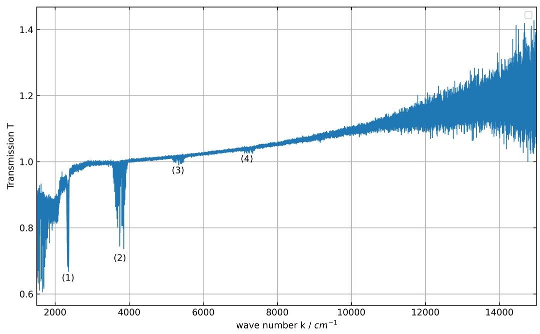

Abbildung 6: Normalized transmission of an empty chamber filled with ambient air. The spectral ranges for gas absorption are highlighted.

To identifythe spectral ranges where the result is influenced by the gas absorption, a measurement at ambient air with a resolution of $0.3\,\mathrm{cm}^{-1}$ and $N = 10$ scans is performed. The measurement at air is normalized. The corresponding reference measurement is done under vacuum. One expects gas absorption dips at frequencies that are similar to the resonance frequencies of the air molecules. The wavenumbers of the measured gas absorption is shown in tab 3 All the measured data agrees with the literature values for the gases H_2O and CO_2.

CO_2	H_2O CO_2	H_2O	H_2O
$2350\,\mathrm{cm}^{-1}$	$3750\,\mathrm{cm}^{-1}$	$5320\,\mathrm{cm}^{-1}$	$7180\,\mathrm{cm}^{-1}$

Tabelle 3: Regions of strong gas absorption in fig 6. The measured values agree with literature values for H_2O and CO_2 [6].

4.2 Signal to noise ratio

5 Bibliography

Literatur

[1] Meschede D., *Gerthsen Physik*, Springer Verlag, 25th Edition, 2018. Modified by Lukas Hein and Elias Schwarzkopf.

[2] Material for F-Praktikum.

[3] Bundesmann C. *Die Bestimmung er optischen Eigenschaften von Halbleitern mit Fourier-Transform-Infrarot-Spektroskopie* https://polariton.exphysik.uni-leipzig.de/subsites/staff/bundesmann/ftir/ifs48.html. Accessed on 2025-04-02.

[4] ??????????????????

[5] ????????

[6] Sun, J. et al. (2024) ‘Tunable diode laser absorption spectroscopy for detection of multi-component gas: a review’, Applied Spectroscopy Reviews, 59(8), pp. 1086–1107. doi: 10.1080/05704928.2024.2302608.